

Analysis and Design Considerations for a Switched Reluctance Motor-Based Drive System for an Electric Bike

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Abstract—Switched reluctance motors (SRMs) are magnet-free, mechanically robust and cost-effective. They also produce no drag torque when unpowered, which makes them attractive for light electric vehicles. This paper presents a systematic analysis and design of an SRM drive for a 250-W, 25-km/h pedelec supplied from a 36-V battery. The vehicle mission is translated into motor torque–speed requirements. We show that the turns must be chosen at a *corner speed* rather than at the assist cut-off speed, and we derive the relation $P_c = k_c V_{dc} I_{pk}$. This relation implies that the peak phase current depends only weakly on machine size. A 12/8 three-phase machine is sized from the output equation, the feasible pole-arc region and the volt-second balance. It is evaluated with a saturable flux-linkage model coupled to a time-domain model of an asymmetric half-bridge converter, and the turn-on angle, turn-off angle and current reference are optimized at every operating point. The resulting $\varnothing 135$ -mm, 3.4-kg (active) design, coupled to an 8:1 planetary gear, delivers the 5.9-N·m launch torque at 31 A. It reaches 75.8% motor efficiency and 72.8% battery-to-shaft efficiency at the rated point, and consumes 5.05 Wh/km on a hilly urban route. A trade study over gear ratio and electric loading quantifies the efficiency–mass–gearing compromise. Converter ratings, the DC-link capacitor, thermal behavior, torque ripple and position sensing are also discussed.

Index Terms—Asymmetric half-bridge converter, electric bicycle, light electric vehicle, machine design, pedelec, switched reluctance motor (SRM), torque ripple.

I. INTRODUCTION

ELECTRIC bicycles are the most widely sold electric vehicles worldwide. Their drives are almost exclusively brushless permanent-magnet (PM) hub or mid-drive motors. PM machines offer high torque density and efficiency, but they rely on rare-earth magnets. Those magnets carry supply-chain and price risk, limit operating temperature and add drag torque when unpowered (cogging and eddy-current losses). This last point matters for a pedelec, which by regulation must be pedaled and is ridden unassisted above 25 km/h [1].

The switched reluctance motor (SRM) addresses these concerns. Its rotor is a stack of laminations, its stator carries concentrated coils, and its phases are electrically and magnetically independent. This gives inherent fault tolerance and zero torque when the windings are not excited [2], [3]. SRMs have been studied extensively for electric and hybrid vehicles [4]–[7]. Their known drawbacks are torque ripple, acoustic noise

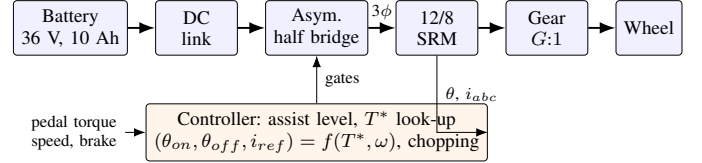


Fig. 1. Architecture of the SRM-based pedelec drive.

and lower efficiency at small power ratings [8], [9]. Classical design procedures [10]–[12] and pole-number studies [13] exist. However, most target kilowatt-to-hundred-kilowatt machines with relatively high DC-link voltages. The 250-W, 36-V pedelec regime brings a different set of constraints: a severe volt-second limit, a launch torque about four times the rated torque, a hub-shaped envelope, and a mission dominated by part-load cruising.

This paper presents a complete analysis and design of an SRM drive for this application, from the vehicle mission to the converter. Its contributions are as follows:

- 1) A mission-to-converter design flow for a pedelec SRM drive, combining analytical sizing with simulation-based current sizing and angle optimization.
- 2) The corner-speed design rule and the invariant $P_c = k_c V_{dc} I_{pk}$. The invariant shows that the peak current is set by the battery voltage and power requirement rather than by machine size, so machine size buys efficiency, not torque.
- 3) Angle-optimized torque–speed, efficiency and loss maps, together with converter, DC-link and thermal ratings, evaluated over a realistic hilly route.
- 4) A trade study over gear ratio and electric loading that gives design guidance on mass and efficiency.

The toolchain is open source, and every number in this paper can be reproduced from it.

II. SYSTEM DESCRIPTION AND DRIVE REQUIREMENTS

A. Drive Architecture

Fig. 1 shows the drive system. A 10-cell (36 V nominal, 30–42 V) Li-ion battery feeds a three-phase asymmetric half-bridge converter through a DC-link capacitor bank. The SRM drives the rear wheel through a single-stage planetary gear of ratio G inside the hub. The controller uses rotor position, phase currents, pedal-torque or cadence sensing and a speed limit to command assistance up to the 25-km/h cut-off.

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The simulation toolchain used in this work is available as open source in the accompanying repository.

TABLE I
MOTOR-SIDE DRIVE REQUIREMENTS ($G = 8$)

Duty point	Condition	Torque (N·m)
Cruise	level, 25 km/h (149 W)	0.88
Rated	250 W at 25 km/h	1.48
Hill	6% grade, 15 km/h	3.03
Launch	8% grade, 0.5 m/s ²	5.93
Base speed n_b	25 km/h	1608 r/min
Corner speed n_c	P_{pk}/T_{pk} , $P_{pk} = 500$ W	805 r/min
Max. speed	1.2 n_b	1930 r/min

B. Road-Load Model

The tractive force at the tire contact patch is

$$F = C_{rr}mg \cos \alpha + mg \sin \alpha + \frac{1}{2}\rho C_d A v^2 + k_m m \dot{v}, \quad (1)$$

where $m = 100$ kg (80-kg rider plus 20-kg bicycle), $C_{rr} = 0.006$, $C_d A = 0.5$ m² (upright rider), $k_m = 1.05$ accounts for rotating inertia, and $\alpha = \arctan(\text{grade})$. With wheel radius $r_w = 0.33$ m and gear efficiency $\eta_g = 0.95$, the motor torque and speed are

$$T_m = \frac{F r_w}{G \eta_g}, \quad \omega_m = \frac{G v}{r_w}. \quad (2)$$

C. Duty Points

Table I lists the motor-side requirements for $G = 8$. Cruising on level ground at 25 km/h needs only 149 W. Launching on an 8% grade at 0.5 m/s² without rider effort, the worst case for throttle or walk-assist use, needs 5.93 N·m. This is four times the rated torque.

D. Corner-Speed Design Rule

In single-pulse operation, the peak flux linkage a phase can reach is limited by the volt-seconds applied during the dwell angle θ_d :

$$\psi_{pk} = \frac{V_{dc} \theta_d}{\omega}. \quad (3)$$

The number of turns per phase follows from $\psi_{pk} = T_{ph} B_{sp} A_p$, where A_p is the stator-pole face area and B_{sp} the design pole flux density. The average torque is $T = (m N_r / 2\pi) W_c$, where W_c is the energy converted per stroke, i.e., the area of the ψ - i loop. With a loop-utilization factor $k_c = W_c / (\psi_{pk} I_{pk})$ and θ_d equal to the stroke angle $\varepsilon = 2\pi / (m N_r)$, the power at the speed ω_c where the turns are chosen is

$$P_c = T_{pk} \omega_c = \frac{m N_r}{2\pi} k_c \psi_{pk} I_{pk} \omega_c = k_c V_{dc} I_{pk}. \quad (4)$$

Equation (4) has two consequences. First, the peak current needed for a given corner power is fixed by the battery voltage, *not by machine size*. A larger machine has fewer turns at the same volt-seconds and therefore saturates at a similar ampere-turn-to-current ratio. Second, choosing the turns for full torque at the 25-km/h base speed would require $P_c = T_{pk} \omega_b \approx 1$ kW and twice the phase current. The launch torque is only needed at walking speed, so the turns are chosen at the corner speed $n_c = P_{pk} / T_{pk} = 805$ r/min (12.5 km/h) with $P_{pk} = 500$ W. Above n_c the drive operates at approximately constant power. The simulations of Section VIII give $k_c = 0.54$ – 0.66 across all nine design variants.

TABLE II
COMPARISON OF POLE CONFIGURATIONS AT 1930 R/MIN

N_s/N_r	m	Stroke	Strokes/rev	f_{ph} (Hz)	Switches
6/4	3	30°	12	129	6
8/6	4	15°	24	193	8
12/8	3	15°	24	257	6
16/12	4	7.5°	48	386	8
24/16	3	7.5°	48	514	6

III. TOPOLOGY SELECTION AND MACHINE DESIGN

A. Pole Configuration

Table II compares candidate configurations at the maximum speed. The 12/8 machine has the same number of strokes per revolution as the four-phase 8/6 but needs only six switches. Its two pole pairs per phase balance the radial forces and shorten the flux paths, which suits the short, large-diameter hub envelope. Its phase frequency (257 Hz) keeps iron losses moderate with 0.35-mm laminations, whereas 16/12 and 24/16 machines [6], [13] double this frequency at no benefit for a 250-W machine. The 6/4 machine has a 30° stroke, which causes large torque dips and makes starting on a grade more difficult.

B. Main Dimensions

In torque form, the SRM output equation [3], [10] reads

$$T = \frac{\pi}{4} k_e k_d k_2 B A D^2 L, \quad (5)$$

where D is the bore diameter and L the stack length. The coefficients are $k_e = 0.85$ (efficiency), $k_d = 0.9$ (duty), $k_2 = 0.7$ ($= 1 - 1/\lambda_u$), $B = 1.7$ T (stator-pole flux density at alignment) and $A = 45$ kA/m (short-term peak electric loading). For $T_{pk} = 5.93$ N·m this gives $D^2 L = 184$ cm³. An aspect ratio $L/D = 0.5$ and a split ratio $D/D_o = 0.53$ give $D = 71.7$ mm, $L = 35.9$ mm and $D_o = 135.3$ mm. We use (5) only as a starting point. Its coefficients are optimistic for small, heavily saturated machines, so the current required to meet T_{pk} is determined by simulation (Section V).

C. Pole Arcs

The stator and rotor pole arcs must satisfy the feasibility conditions of Lawrenson *et al.* [14]:

$$\min(\beta_s, \beta_r) \geq \varepsilon, \quad \beta_s \leq \beta_r, \quad \beta_s + \beta_r < \frac{2\pi}{N_r}. \quad (6)$$

The first condition guarantees self-starting from every rotor position. The second preserves slot area. The third creates a zero-overlap region that keeps the unaligned inductance low. We select $\beta_s = 15^\circ$ and $\beta_r = 17^\circ$, close to the lower-left vertex of the feasible region in Fig. 2(b). This choice maximizes the inductance ratio and slot area. The 2° excess of β_r creates a short dwell before alignment in which the tail current can decay before negative torque is produced.

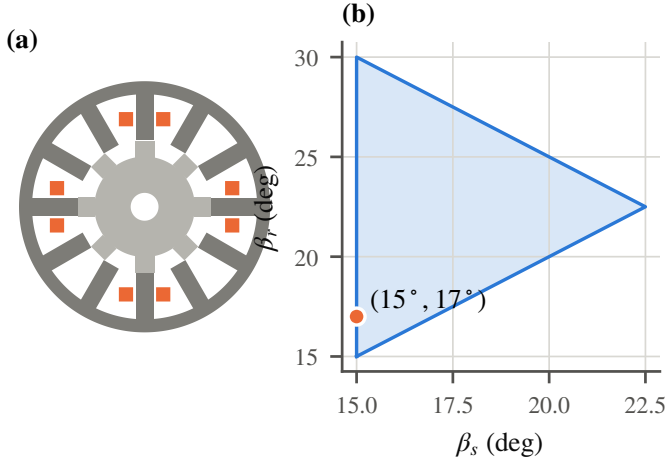


Fig. 2. (a) Lamination of the designed 12/8 SRM with phase-A coil sides marked. (b) Feasible pole-arc region (6) and the selected point.

TABLE III
MAIN PARAMETERS OF THE DESIGNED 12/8 SRM

Outer diameter	135.3 mm	Turns/pole	41
Bore diameter	71.7 mm	Wire diameter	1.66 mm
Stack length	35.9 mm	R_{ph} (100 °C)	0.198 Ω
Air gap	0.30 mm	L_a (unsat.)	9.06 mH
β_s / β_r	15° / 17°	L_u	1.37 mH
Stator pole width	9.36 mm	L_a / L_u	6.6
Stator pole height	24.8 mm	ψ_k	85.3 mWb
Rotor pole height	9.0 mm	Iron mass	2.31 kg
Stator/rotor yoke	7.0/19.1 mm	Copper mass	1.09 kg
Slot area	394 mm ²	Rotor inertia	4.1×10^{-4} kg·m ²

D. Poles, Yokes, Turns and Winding

The stator yoke is $y_s = 0.75 w_s$, where $w_s = 9.4$ mm is the stator-pole width. The pole flux divides in two in the yoke, so $0.5 w_s$ is the minimum. The extra margin limits yoke saturation and stiffens the stator against the ovalizing vibration mode, which is the dominant source of SRM noise [2]. The rotor-pole height is limited to $30g$ ($g = 0.3$ mm) as a compromise between unaligned inductance and pole stiffness.

With (3) at n_c , the minimum battery voltage of 30 V and a one-stroke dwell, $\psi_{pk} = 93.2$ mWb. This gives $T_{ph} = 164$ turns, i.e., 41 turns on each of the four poles of a phase. Bobbin-wound concentrated coils with a net fill factor of 0.45 use 1.66-mm wire. The mean turn length is 115 mm and $R_{ph} = 0.198 \Omega$ at 100 °C. Table III summarizes the design.

E. Inductances

The aligned unsaturated inductance follows from two air gaps in series per pole pair, with all $p = N_s/m$ poles of a phase in series:

$$L_a = k_{fe} \frac{\mu_0 (w_s + 2g) L T_{ph}^2}{p g}, \quad (7)$$

where $k_{fe} = 0.9$ accounts for iron MMF and the $2g$ term for fringing. In the unaligned position, the flux crosses into the

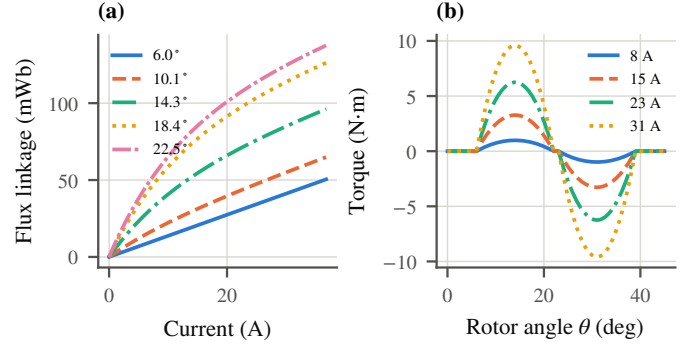


Fig. 3. (a) Flux linkage versus current from the onset of pole overlap (6.0°) to the aligned (22.5°) position. (b) Static torque of one phase.

rotor slot (path length $g + h_r$) in parallel with two side paths to the neighboring rotor-pole corners (path length l_s). This gives

$$\frac{L_u}{L_a} = \frac{k_{ew}}{k_{fe}} \left(\frac{g}{g + h_r} + \frac{g}{l_s} \right), \quad (8)$$

where $k_{ew} = 1.35$ covers end-winding and slot leakage. The result, $L_a/L_u = 6.6$, is typical of small 12/8 machines with this air gap.

IV. DRIVE MODELING

A. Saturable Flux-Linkage Model

The aligned magnetization curve is represented by an exponential knee model in the spirit of [15], [16]:

$$\psi_a(i) = L_{sat} i + \psi_k \left(1 - e^{-(L_a - L_{sat})i/\psi_k} \right), \quad (9)$$

whose slope is L_a at $i = 0$ and L_{sat} in deep saturation. The knee flux is $\psi_k = B_k A_p T_{ph}$ with $B_k = 1.55$ T. A position function $w(\theta) \in [0, 1]$ blends (9) with the linear unaligned characteristic:

$$\psi(i, \theta) = L_u i + w(\theta) [\psi_a(i) - L_u i]. \quad (10)$$

Here w is unity while the stator pole is fully covered and zero without overlap. Between these, it follows a C^1 smooth step that is widened by a fringing angle of $2g/r$. Because w multiplies a function of i alone, the co-energy and torque have closed forms:

$$W'(i, \theta) = \frac{1}{2} L_u i^2 + w(\theta) [W'_a(i) - \frac{1}{2} L_u i^2], \quad (11)$$

$$T(i, \theta) = \frac{dw}{d\theta} \left[W'_a(i) - \frac{1}{2} L_u i^2 \right], \quad (12)$$

with $W'_a(i) = \int_0^i \psi_a di$. Fig. 3 shows the magnetization curves and the static torque. Torque is concentrated in the overlap region and grows sub-quadratically with current once the poles saturate.

B. Converter and Circuit Model

Each phase obeys $d\psi/dt = v - R_{ph}i$, with the converter imposing

$$v = \begin{cases} V_{dc} - 2R_{ds}i & \text{both switches on (magnetize)} \\ -(R_{ds}i + V_D) & \text{one switch on (freewheel)} \\ -(V_{dc} + 2V_D) & \text{both off, } i > 0 \text{ (demagnetize)} \end{cases} \quad (13)$$

with $R_{ds} = 4 \text{ m}\Omega$ and $V_D = 0.8 \text{ V}$. The flux equation is integrated at 40 kHz or at least 1800 steps per period, with i obtained from (10) by Newton iteration. A sampled hysteresis controller with a 2% band regulates the current by soft chopping. At high speed the back-EMF prevents the reference from being reached, and single-pulse operation follows naturally. Phases are identical and displaced by ε , so one phase is simulated to periodic steady state and the total torque is

$$T_e(\theta) = \sum_{k=0}^{m-1} T_{ph}(\theta - k\varepsilon). \quad (14)$$

The DC-link current is reconstructed in the same way. As a verification, the DC input power equals the sum of electromagnetic power, copper loss and conduction loss to within 1.1% at all speeds from 80 to 1900 r/min.

C. Loss Models

Copper loss is $mR_{ph}I_{rms}^2$. Converter conduction loss is integrated from (13), and switching loss is $\frac{1}{2}V_{dc}I t_{sw}f_{sw}$ per transition with $t_{sw} = 60 \text{ ns}$. Iron loss in the stator poles, stator yoke and rotor is computed with the Bertotti form [17]

$$p_{fe} = k_{ns} (k_h f B^2 + k_e (fB)^2) \quad (\text{W/kg}), \quad (15)$$

calibrated to 2.7 W/kg at 1.5 T/50 Hz (M270-35A). Here $k_{ns} = 1.3$ accounts for the unipolar, non-sinusoidal SRM flux, $f = N_r n / 60$ in the stator and $N_s n / 120$ in the rotor. Bearing and windage losses are below 2 W.

V. CONTROL AND DESIGN FLOW

A. Commutation-Angle Optimization

The turn-on angle must lead the start of pole overlap θ_{ov} by roughly the angle needed to build up the reference current in L_u :

$$\theta_{adv} \approx \frac{\omega L_u i_{ref}}{V_{dc} - (R_{ph} + 2R_{ds})i_{ref}}. \quad (16)$$

At each speed–torque point (ω, T^*) , we solve

$$\begin{aligned} & \min_{\theta_{on}, \theta_{off}, i_{ref}} P_{in} \\ & \text{s.t. } \bar{T}_e = T^*, \quad i_{ref} \leq I_{max}, \quad \theta_{off} - \theta_{on} \leq 0.55 \tau_r, \end{aligned} \quad (17)$$

where $\tau_r = 2\pi/N_r$. The search covers $\theta_{on} \in [\theta_{ov} - 1.6\theta_{adv}, \theta_{ov}]$ and θ_{off} at 45–95% of the interval from overlap to alignment. For each angle pair, i_{ref} is found with Brent's method. The dwell constraint is essential. Without it, the optimizer drifts into continuous conduction at high speed, and the tail current then produces negative torque after alignment. In a controller, the solution becomes a two-dimensional look-up table $(\theta_{on}, \theta_{off}, i_{ref}) = f(T^*, \omega)$, as in [5], [18].

B. Design Flow

Fig. 4 summarizes the procedure. The analytical machine design is followed by a simulation step that finds the smallest current giving T_{pk} with optimal angles. Here this current is 27.9 A. The converter limit is set 10% higher, at $I_{max} = 30.7 \text{ A}$.

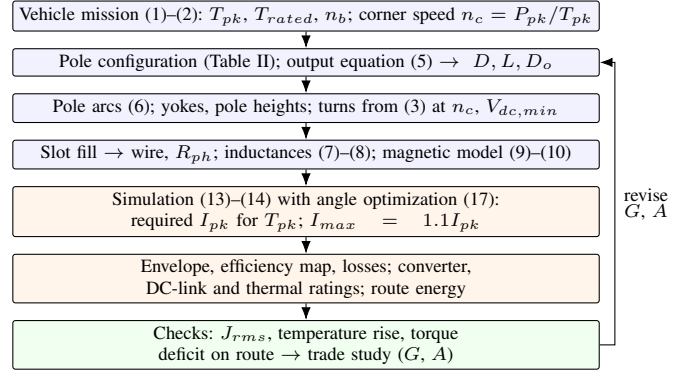


Fig. 4. Mission-to-converter design flow of the SRM e-bike drive.

VI. RESULTS

A. Current and Torque Waveforms

Fig. 5 shows the steady-state waveforms in the two operating modes. At launch (80 r/min), soft chopping holds the current at 30.7 A. The pole saturates ($B_{sp} \approx 2 \text{ T}$), and late commutation ($\theta_{off} = 21.7^\circ$) uses most of the stroke, giving a mean torque of 6.81 N·m. At the rated point (1608 r/min), the turn-on angle is advanced to -4.9° . The current peaks early, and the flux rises linearly to 57 mWb at turn-off. The peak-to-peak torque ripple is 113% at launch and 148% at the rated point. This is inherent to a three-phase machine with $\beta_s = \varepsilon$ under current-reference control, and it is discussed in Section IX.

B. Torque–Speed Capability

At $I_{max} = 30.7 \text{ A}$, the envelope in Fig. 6 delivers 6.8 N·m at low speed and about 3.6 N·m at 25 km/h. The available power exceeds the 500-W target above about 900 r/min. All duty points of Table I lie inside the envelope. A steady 8% climb at more than about 20 km/h without rider input lies outside it. This is consistent with the pedelec concept, in which the rider contributes on steep grades.

C. Efficiency and Losses

Fig. 7 shows the battery-to-shaft efficiency map with the operating points of the route of Section VII overlaid. The maximum system efficiency is 78.6% and the maximum motor efficiency 81.4%. Both occur at medium torque near and above the base speed, which is where the route's cruise and rated points lie. Table IV and Fig. 8 give the operating points and their loss breakdown. Copper loss dominates everywhere. This is the characteristic SRM penalty: the stator also carries the magnetizing current, as unipolar pulses with a high RMS-to-average ratio. At the rated point, iron loss accounts for about 21% and converter conduction loss for about 14% of the 86-W total. The diode drops are significant at only 36 V.

D. Converter Design

The asymmetric half bridge (two switches and two diodes per phase) is the only common SRM converter that provides $+V_{dc}$, 0 and $-V_{dc}$ independently in every phase. These

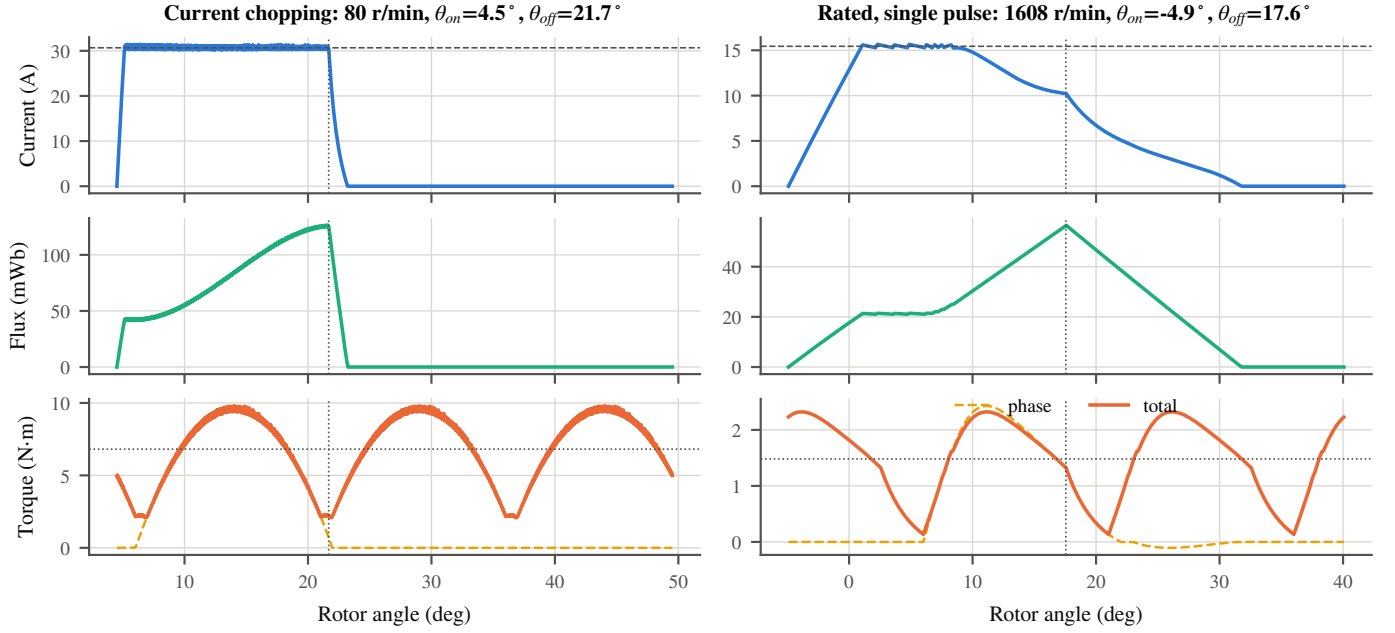


Fig. 5. Steady-state phase current, flux linkage and torque (single phase and total) at launch (left, current chopping) and at the rated point (right, single pulse). Dotted vertical lines mark θ_{off} .

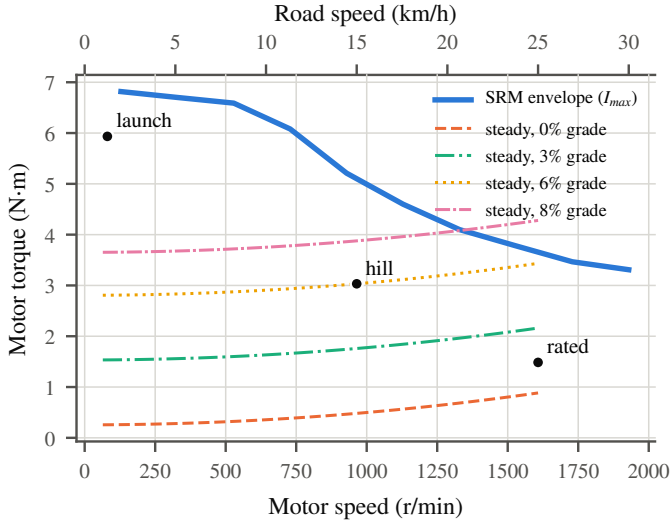


Fig. 6. Torque envelope at I_{max} with optimized angles, steady road-load torque for 0–8% grades, and the duty points of Table I.

TABLE IV
OPTIMIZED OPERATING POINTS

Point	n r/min	T_e N·m	θ_{on} deg	θ_{off} deg	I_{rms} A	η_{mot} %	η_{sys} %
Launch	80	6.81	4.5	21.7	18.9	20.7	18.8
Hill 6%	965	3.03	-0.5	17.6	12.1	73.4	70.8
Rated	1608	1.48	-4.9	17.6	9.5	75.8	72.8
Cruise	1608	0.88	-4.9	17.6	6.8	75.6	71.6

states allow fast demagnetization, soft chopping and phase-independent fault handling [2], [3]. The $(n+1)$ -switch and Miller converters share a switch, which prevents phase overlap at speed and forces hard chopping. The C-dump converter

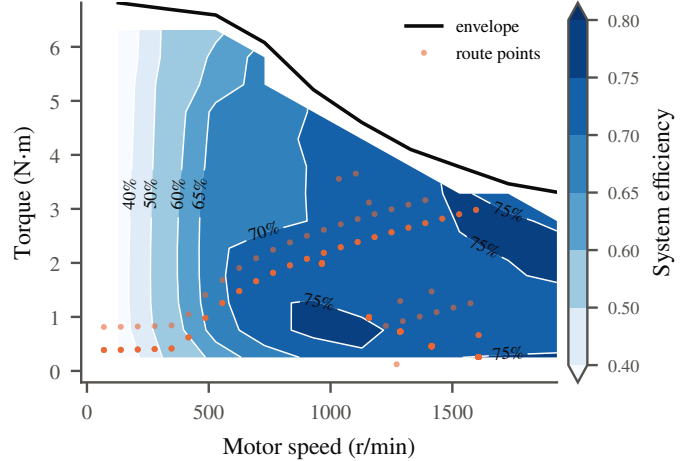


Fig. 7. Battery-to-shaft efficiency map (optimized angles at each point), with the torque envelope and the operating points of the route simulation.

needs an extra inductor and capacitor, and the split-DC converter halves the phase voltage and needs an even number of phases. Table V lists the ratings. The DC-link capacitor is dimensioned by ripple current rather than by capacitance. The SRM draws unipolar pulses, and during demagnetization the energy returns to the link. The resulting ripple current exceeds the average battery current. The average hysteresis switching frequency is 2.4–6 kHz, which is audible. A production design should use fixed-frequency (≥ 16 kHz) PWM current control with the same soft-chopping states.

E. Thermal Assessment

At the continuous rated point the motor dissipates 73 W. The current density is 4.4 A/mm². The hub shell provides 0.075 m²

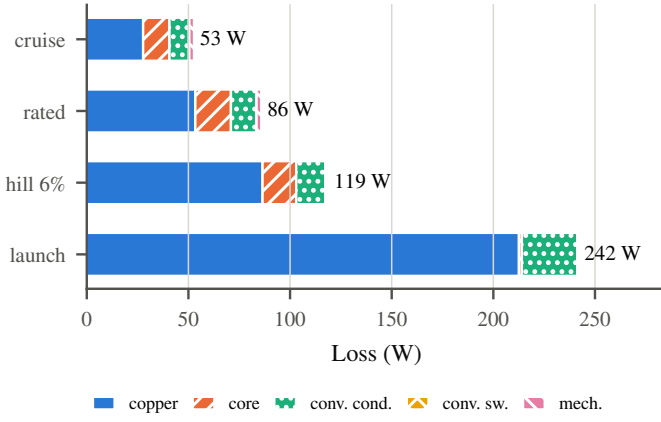


Fig. 8. Loss breakdown at the operating points of Table IV.

TABLE V
CONVERTER AND DC-LINK RATINGS

Device voltage ($1.5 V_{bat,max}$) / selected	63 V / 100 V MOSFET
Phase current limit I_{max}	30.7 A
Device RMS current at launch	18.9 A
Average battery current, rated / launch	8.7 A / 8.2 A
DC-link ripple current (RMS), rated / launch	10.9 A / 15.4 A
DC-link capacitance for $\leq 5\%$ voltage ripple	3.8 mF

of cooling surface, and riding gives a convection coefficient of about $25 \text{ W/m}^2\text{K}$. A lumped model then predicts a temperature rise of 39 K, or a winding hot spot of about 77°C at 30°C ambient. This leaves a wide margin to class-F insulation for sustained climbing. The launch point (240 W loss) is limited to tens of seconds by the thermal capacity of the copper (about 420 J/K). Because the rotor contains no magnets, the machine tolerates these transients without risk of demagnetization.

VII. ROUTE ENERGY EVALUATION

A 6.4-km urban route with five stops, a 700-m climb at 6% and a 900-m descent at -4% was simulated quasi-statically (Fig. 9). The rider supplies up to 100 W, force-limited at low speed. The motor supplies the remaining tractive power below 25 km/h, with efficiency interpolated from Fig. 7, and braking is mechanical. The ride takes 18.9 min. The rider contributes 25.5 Wh and the motor 23.4 Wh at the shaft, which draws 32.3 Wh from the battery. The cycle-average drive efficiency is therefore 72.3%. Consumption is 5.05 Wh/km, which corresponds to about 64 km of range from 90% of a 360-Wh pack. The torque demand never exceeded the envelope. Launches briefly reach about 3.7 N·m and 660 W of battery power.

VIII. DESIGN TRADE-OFFS

Table VI and Fig. 10 vary the gear ratio G and the peak electric loading A , redesigning the machine and repeating the current sizing each time. Three observations stand out.

- 1) The required peak current varies only from 27.7 to 33.7 A across a fourfold range of active mass, as predicted by (4).

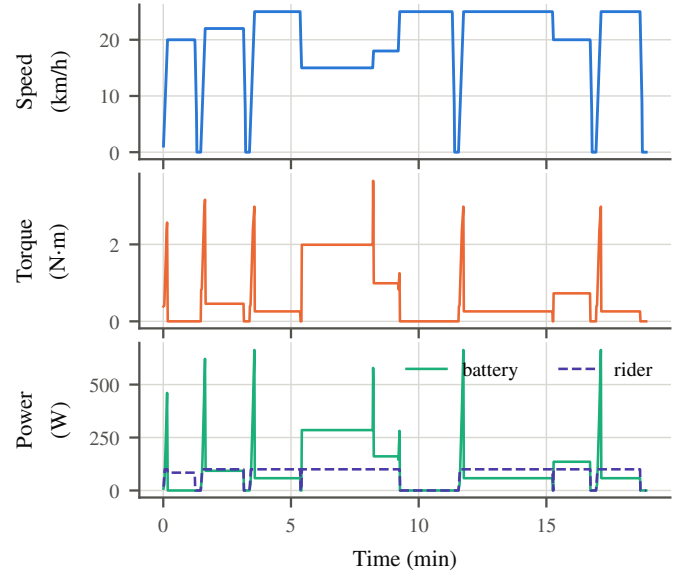


Fig. 9. Route simulation: speed, motor torque, and battery and rider power.

TABLE VI
TRADE STUDY: GEAR RATIO G AND PEAK ELECTRIC LOADING A

G	A kA/m	D_o mm	L mm	Mass kg	I_{max} A	η_{mot} %	η_{sys} %
5	60	143.8	38.1	4.10	27.7	70.6	68.5
5	45	158.3	41.9	5.51	29.2	76.3	73.6
5	35	172.1	45.6	7.14	31.2	81.0	78.0
8	60	122.9	32.6	2.53	29.3	70.6	68.1
8	45	135.3	35.9	3.40	30.7	75.8	72.8
8	35	147.1	39.0	4.40	32.0	79.9	76.6
11	60	110.6	29.3	1.82	30.3	69.9	67.3
11	45	121.7	32.2	2.45	31.6	74.8	71.6
11	35	132.3	35.1	3.17	33.7	80.2	77.0

- 2) Efficiency is governed mainly by electric loading. Each step from 60 to 45 to 35 kA/m adds about five points of rated efficiency for about 30% more mass.
- 3) A higher gear ratio reduces machine size. The variant with $G = 11$ and $A = 35$ kA/m has the baseline's diameter (132 mm) and slightly lower mass (3.2 kg), yet gains about four points of system efficiency. The cost is a two-stage gear and a higher iron-loss frequency.

The single-stage $G = 8$, $A = 45$ kA/m baseline favors cost and simplicity. The $G = 11$, $A = 35$ kA/m variant is preferable when range is the priority.

IX. DISCUSSION

A. Torque Ripple and Acoustic Noise

With $\beta_s = \varepsilon$ and flat-top current references, essentially one phase conducts at a time, which explains the 110–150% ripple. The gear and wheel inertia filter much of it. On a bicycle frame, however, the residual is felt as vibration and heard as noise, together with the radial-force excitation of the stator. Mitigation options, in increasing order of cost, are as follows. Torque-sharing functions overlap the incoming and outgoing phases and typically reduce ripple to a few tens of percent at low speed for a modest increase in RMS current [8], [9], [19].

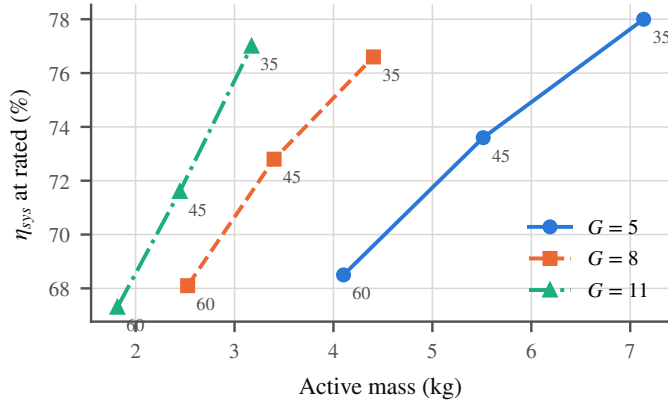


Fig. 10. Rated-point system efficiency versus active mass for gear ratios 5, 8 and 11. The labels give the peak electric loading in kA/m.

Wider rotor poles increase phase overlap. Shaped or notched poles help further. On the mechanical side, a stiff, round stator yoke and a compliant gear coupling reduce noise. The look-up-table structure of Section V accommodates torque-sharing references directly.

B. Position Sensing

Commutation needs rotor position to about 1° mechanical. Three Hall sensors on a target ring provide 15° edges, which can be interpolated using speed. Sensorless schemes based on flux observers or inductance probing [20] are attractive for sealed hubs. They must, however, achieve reliable stand-still starts on a grade, which is exactly the launch condition that sizes the drive.

C. Mechanical Considerations

The 0.3-mm air gap is small for a wheel hub that sees road shocks. It requires a stiff axle, preloaded bearings on both sides of the stack and careful tolerance analysis. Every additional 0.05 mm reduces L_a by about 15%. The magnet-free rotor (4.1×10^{-4} kg·m²) is robust at overspeed and can be sealed without corrosion concerns. Because no drag torque exists when unpowered, the freewheel clutch normally found in geared hubs becomes optional.

D. Comparison With PM Hub Drives and Limitations

Commercial PM geared hubs of similar rating typically quote peak efficiencies in the low-to-mid 80% range. The SRM baseline trades several points of efficiency for the absence of magnets and drag torque, lower material cost and fault tolerance. The trade study shows that most of this gap can be closed by mass or gearing. Further gains are possible from thinner (0.2-mm) laminations, a 0.25-mm air gap and segmented stators with fill factors above 0.5.

The results rest on analytical permeance estimates of L_a , L_u and the knee flux, on a lumped iron-loss model and on a one-node thermal model. We estimate their uncertainty at about $\pm 15\%$ for inductance and torque and $\pm 30\%$ for iron loss. Mutual coupling between phases is neglected, which

is justified for short-pitched concentrated coils. The flux-linkage model exposes the same interface as a tabulated $\psi(i, \theta)$ characteristic, so finite-element data can replace it directly. Finite-element verification and prototype measurements are the next steps of this work.

X. CONCLUSION

This paper has presented the analysis and design of a switched reluctance motor drive for a 250-W pedelec, from vehicle mission to converter. The key design insight is the corner-speed rule. Choosing the turns at $n_c = P_{pk}/T_{pk}$ rather than at the 25-km/h base speed halves the launch current. The invariant $P_c = k_c V_{dc} I_{pk}$, with $k_c \approx 0.54\text{--}0.66$, shows that peak current is set by battery voltage and power rather than by machine size. The resulting 12/8 SRM has an outer diameter of 135 mm and 3.4 kg of active mass, and drives the wheel through an 8:1 gear from a three-phase asymmetric half bridge with 100-V MOSFETs. It meets all duty points, including a 5.9-N·m launch on an 8% grade at 31 A. At the rated point it reaches 75.8% motor and 72.8% system efficiency, and it consumes 5.05 Wh/km on a hilly urban route, which corresponds to about 64 km of range. With higher gearing and lower electric loading, efficiency rises to about 80% at equal mass. These results indicate that the SRM is a technically viable, magnet-free alternative for pedelec drives, provided that torque ripple is handled by torque-sharing control and careful mechanical design.

REFERENCES

- [1] European Committee for Standardization, "EN 15194:2017 — cycles — electrically power assisted cycles — EPAC bicycles," CEN, Brussels, Belgium, 2017.
- [2] T. J. E. Miller, *Switched Reluctance Motors and Their Control*. Hillsboro, OH, USA and Oxford, U.K.: Magna Physics Publishing and Clarendon Press, 1993.
- [3] R. Krishnan, *Switched Reluctance Motor Drives: Modeling, Simulation, Analysis, Design, and Applications*. Boca Raton, FL, USA: CRC Press, 2001.
- [4] K. M. Rahman, B. Fahimi, G. Suresh, A. V. Rajarathnam, and M. Ehsani, "Advantages of switched reluctance motor applications to EV and HEV: Design and control issues," *IEEE Trans. Ind. Appl.*, vol. 36, no. 1, pp. 111–121, 2000.
- [5] X. D. Xue, K. W. E. Cheng, J. K. Lin, Z. Zhang, K. F. Luk, T. W. Ng, and N. C. Cheung, "Optimal control method of motoring operation for SRM drives in electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 59, no. 3, pp. 1191–1204, 2010.
- [6] J. W. Jiang, B. Bilgin, and A. Emadi, "Three-phase 24/16 switched reluctance machine for a hybrid electric powertrain," *IEEE Trans. Transport. Electric.*, vol. 3, no. 1, pp. 76–85, 2017.
- [7] B. Bilgin, J. W. Jiang, and A. Emadi, Eds., *Switched Reluctance Motor Drives: Fundamentals to Applications*. Boca Raton, FL, USA: CRC Press, 2019.
- [8] I. Husain, "Minimization of torque ripple in SRM drives," *IEEE Trans. Ind. Electron.*, vol. 49, no. 1, pp. 28–39, 2002.
- [9] V. P. Vujčić, "Minimization of torque ripple and copper losses in switched reluctance drive," *IEEE Trans. Power Electron.*, vol. 27, no. 1, pp. 388–399, 2012.
- [10] R. Krishnan, R. Arumugam, and J. F. Lindsay, "Design procedure for switched-reluctance motors," *IEEE Trans. Ind. Appl.*, vol. 24, no. 3, pp. 456–461, 1988.
- [11] A. V. Radun, "Design considerations for the switched reluctance motor," *IEEE Trans. Ind. Appl.*, vol. 31, no. 5, pp. 1079–1087, 1995.
- [12] K. Vijayakumar, R. Karthikeyan, S. Paramasivam, R. Arumugam, and K. N. Srinivas, "Switched reluctance motor modeling, design, simulation, and analysis: A comprehensive review," *IEEE Trans. Magn.*, vol. 44, no. 12, pp. 4605–4617, 2008.

- [13] B. Bilgin, A. Emadi, and M. Krishnamurthy, "Design considerations for switched reluctance machines with a higher number of rotor poles," *IEEE Trans. Ind. Electron.*, vol. 59, no. 10, pp. 3745–3756, 2012.
- [14] P. J. Lawrenson, J. M. Stephenson, P. T. Blenkinsop, J. Corda, and N. N. Fulton, "Variable-speed switched reluctance motors," *IEE Proc. B, Electr. Power Appl.*, vol. 127, no. 4, pp. 253–265, 1980.
- [15] D. A. Torrey and J. H. Lang, "Modelling a nonlinear variable-reluctance motor drive," *IEE Proc. B, Electr. Power Appl.*, vol. 137, no. 5, pp. 314–326, 1990.
- [16] M. Ilic-Spong, R. Marino, S. M. Peresada, and D. G. Taylor, "Feedback linearizing control of switched reluctance motors," *IEEE Trans. Autom. Control*, vol. 32, no. 5, pp. 371–379, 1987.
- [17] G. Bertotti, "General properties of power losses in soft ferromagnetic materials," *IEEE Trans. Magn.*, vol. 24, no. 1, pp. 621–630, 1988.
- [18] P. C. Kjaer, J. J. Gribble, and T. J. E. Miller, "High-grade control of switched reluctance machines," *IEEE Trans. Ind. Appl.*, vol. 33, no. 6, pp. 1585–1593, 1997.
- [19] J. Ye, B. Bilgin, and A. Emadi, "An extended-speed low-ripple torque control of switched reluctance motor drives," *IEEE Trans. Power Electron.*, vol. 30, no. 3, pp. 1457–1470, 2015.
- [20] M. Ehsani and B. Fahimi, "Elimination of position sensors in switched reluctance motor drives: State of the art and future trends," *IEEE Trans. Ind. Electron.*, vol. 49, no. 1, pp. 40–47, 2002.